5.1 Fully Reproducible System

5.1.1 Hardware Requirements

Component	Specification
Robotics Platform	LEGO Mindstorms EV3 Education Set 45544
Driving Motor	EV3 Medium Motor
Steering Motor	EV3 Medium Motor
Vision Sensor	Pixy Camera
Distance Sensors	2 × Ultrasonic Sensors
Orientation Sensor	EV3 Gyro Sensor
Colour Sensor	EV3 Color Sensor
MUX	EV3 Sensor Multiplexer
Power Supply	6 × 1.5 V AA Batteries
Chassis	Custom 3D-Printed Chassis

5.1.2 Port Configuration

EV3 Port	Connected Component	Notes
Port A	Steering Motor	-
Port D	Driving Motor	-
Port 1	Sensor MUX	Channel 1 Left Ultrasonic Sensor Channel 2 Right Ultrasonic Sensor
Port 2	Gyro Sensor	-
Port 3	Pixy Camera	-
Port 4	Color Sensor	-

Refer to Wiring Diagram

5.1.3 Software Requirements

Software	Function
Clev3r	• Used for robot programming, developing and executing the BP Program. • Refer to installation address:https://rbtx.ru/clever/
PixyMon	• Used to configure and tune Pixy Camera settings, including colour signatures and detection parameters. • Refer to installation address:https://pixycam.com/downloads-pixy2/
Any 3D Printing Software	• Used to prepare the provided 3D model files (STL) for printing. • 3D Printing Part

5.1.4 Setup Instructions

Build the Robot	• Assemble the robot according to the provided CAD files and mechanical design documentation. • Connect all motors and sensors to their designated EV3 ports by referring to the wiring diagram.
Install the Required Software	• Download and install Clev3R for robot programming. • Download and install PixyMon for Pixy Camera configuration.
Configure and Tune the Pixy Camera	• Connect the Pixy Camera to the computer. • Configure the colour signatures and adjust the camera settings to ensure reliable pillar detection. • Refer to link for tutorial of color recognization https://youtu.be/fSxTpedWmcY
Load and Run the Provided Program	• Locate the provided BP Program in the GitHub Repository. • Load and execute the program using the Clev3R environment. • Refer to link for Clev3r's tutorial https://youtu.be/DMhK63uLRww
Perform Additional Tuning and Testing	• Make minor adjustments to the robot's parameters according to your hardware configuration and testing environment. • Refer to the Algorithm section for explanations of the relevant parameters and their functions. • Conduct your own testing and collect performance data to determine the most suitable settings for your robot.

- The BP Program provided in the GitHub Repository serves as the final program for operating the Generation 4 robot.
- Users can access the program and supporting documentation to reproduce the robot's configuration and operation.
- After completing the hardware assembly and software setup, users can load and execute the BP Program using the Clev3R environment.
- The robot can then be tested using the documented testing procedures to verify its autonomous navigation, pillar detection, obstacle avoidance and parking performance.